THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS CHAPTER TEST SET B

Chapter 13: **Algebra Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Think of a number n. Add 3, multiply by 4, then subtract 12. The result is: 1

(a) $4n + 3$ **(b)** $3n$ **(c)** $4n$ **(d)** $n + 3$

2. Think of a number n. Add 8, multiply by 3, then subtract 24. The result is: 1

(a) $3n$ **(b)** $8n$ **(c)** $3n + 8$ **(d)** $n + 8$

3. The two-digit number with tens digit 6 and ones digit 4 can be written as: 1

(a) $10 \times 4 + 6$ **(b)** $6 + 4$ **(c)** 6×4 **(d)** $10 \times 6 + 4$

4. In a number pyramid, each block is the sum of the two below it. The bottom row is 4, 7, 2. The top block is: 1

(a) 13 **(b)** 22 **(c)** 27 **(d)** 20

5. The algebraic expression for 'five more than twice a number x' is: 1

(a) $2(x + 5)$ **(b)** $x + 10$ **(c)** $5x + 2$ **(d)** $2x + 5$

6. Three consecutive even numbers starting with $2k$ are: 1

(a) $2k$, $3k$, $4k$ **(b)** $2k$, $2k + 1$, $2k + 2$

(c) k, $k + 2$, $k + 4$ **(d)** $2k$, $2k + 2$, $2k + 4$

7. The number $100a + 10b + c$ has digits: 1

(a) 100, 10, 1 **(b)** a, b, c

(c) abc as product **(d)** $a + b$, c

8. In a calendar, the dates in a 2×2 block starting at 12 are 12, 13, 19, 20. Their sum is: 1

(a) 66 **(b)** 62 **(c)** 56 **(d)** 64

9. If n is any integer, $n^2 + n$ is always: 1

(a) odd **(b)** even **(c)** a square **(d)** prime

10. Which algebraic fact explains why $a \times b = b \times a$? 1

(a) commutativity **(b)** associativity **(c)** distributivity **(d)** closure

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Show that the sum of three consecutive numbers is divisible by 3. 2

12. Find the largest product of two numbers whose sum is 30. 2

13. Show that the difference of squares of consecutive numbers is odd. 2

14. In a 2×2 block of a calendar starting at x, write the sum of the four dates. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. A 2×2 block of a calendar has top-left date 4. Find the four dates, their sum, and show sum $= 4k + 16$. 3

16. A 2×2 block of a calendar has top-left date 7. Find the four dates, their sum, and show sum $= 4k + 16$. 3

17. A 2×2 block of a calendar has top-left date 9. Find the four dates, their sum, and show sum $= 4k + 16$. 3

OR

A 2×2 block of a calendar has top-left date 12. Find the four dates, their sum, and show sum $= 4k + 16$. 3

18. A 2×2 block of a calendar has top-left date 15. Find the four dates, their sum, and show sum $= 4k + 16$. 3

OR

A 2×2 block of a calendar has top-left date 21. Find the four dates, their sum, and show sum $= 4k + 16$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. **Case study:** A fence of 40 m is used to make a rectangular pen against a wall (three sides). 5

(i) Write area in terms of width w

(ii) Find w for maximum area.

(iii) Find the maximum area.

OR

(i) Show that $\overline{abc} - \overline{cba}$ is a multiple of 99. 5

(ii) Verify with 721.

20. (i) Prove that the sum of 5 consecutive integers is divisible by 5. 5

(ii) Find the five numbers with sum 85.

OR

Case study: Anu says: pick a 3-digit number, write it twice to get a 6-digit number; it is divisible by 7, 11, 13. 5

(i) Test with 345.

(ii) Explain algebraically.

(iii) Test another.

END OF QUESTION PAPER — Total: 40 Marks